



www.gethelium.co

AI Product Engineer — Take-Home Assignment

Build an AI-Native Social Media Automation MVP for a D2C Brand

Estimated Time: 60–90 minutes

Submission: Working MVP + short explanation/demo

AI Tools: Encouraged

The Problem:

D2C brands need to consistently create social media content, but a large amount of time goes into deciding:

- What content should we create?
- Who should we create it for?
- When should we publish it?

Today, this process is often manual and disconnected from product data, audience insights, and previous content performance.

Your Challenge:

Build a small AI-powered MVP that helps a D2C brand make these decisions and then automates part of the content creation workflow.

The product should ideally go beyond simply generating captions. It should have some logic for deciding what content is worth creating and why.

What You Can Build:

You can decide the exact workflow, but a possible flow could be:

Product / Brand Information

↓

AI recommends content opportunities

↓

Select audience + objective

↓

Generate platform-specific content

↓

Edit / approve

↓

Schedule or add to a content calendar

You do not need to build the entire flow.

Focus on building a small, polished MVP that demonstrates your product and engineering judgment.

Inputs:

Assume the brand provides:

- Brand description and tone of voice
- Product information
- Target audience
- A few examples of previous social posts
- Basic performance information for previous posts

You may create sample/mock data for the assignment.

We Want to See:

1. Product Thinking

How did you decide:

- What the system should automate?
- What content it should recommend?
- Who the content is for?
- When it should be published?

2. AI-Native Approach

Use AI meaningfully in the product.

You may use LLMs, agents, RAG, image generation, or other AI tools where appropriate.

3. Working Product

Build a functional MVP rather than just a UI mockup.

The implementation can be lightweight, but we should be able to understand and experience the core workflow.

4. User Experience

The product should be simple enough for a D2C marketer to use without technical knowledge.

Submission:

Please share:

1. Working MVP / deployed link or runnable project
2. GitHub repository
3. A short README or 1–2 page explanation covering:
 - Problem you chose to solve
 - Your product decisions
 - How AI is being used
 - What you would build next
4. 2–3 minute demo video (*optional but preferred*)

You are encouraged to use AI coding tools. Please mention the tools you used and briefly explain how you used them.

Evaluation:

We will primarily evaluate:

- Product thinking & problem selection
- AI implementation
- Engineering quality
- UX and simplicity
- Customer empathy
- Speed of execution and ability to make thoughtful trade-offs

You are not being evaluated on the number of features you build. A small, well-thought-out product is better than a large, unfinished one.